Primitive roots modulo primes $2^k\ell + 1$, formalised in Lean 4: Chebyshev's base-10 criterion and the least primitive root of a safe prime

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Version 2 adds the Layer-14 and Layer-15 checks (Section 7.1)

Abstract

We formalise, in Lean 4 with Mathlib, two classical elementary results about primitive roots modulo primes of a special shape. The first is Chebyshev's criterion: if p and $q = 4p + 1$ are prime and $p \equiv 2 \pmod 5$, then 10 is a primitive root modulo q. We also formalise the converse, which follows from the same argument; we have not found it stated explicitly in the literature. The second is the theorem of Ramesh and Makeshwari (2022) that the least positive primitive root modulo a safe prime $p = 2q+1$ is itself prime. Here the formal statement needs no separate existence hypothesis: we also prove that the least positive primitive root exists, using Mathlib's cyclicity of $(\mathbb{Z}/p\mathbb{Z})^\times$. Both proofs reduce to the same criterion for primes $p = 2^k\ell + 1$ with ℓ an odd prime: a nonzero residue is a primitive root exactly when it is a quadratic nonresidue and its 2^k-th power is not 1. We formalise this criterion for all $k \geq 1$, show that the two criteria above are equivalent, via Euler's criterion, to its cases $k = 2$ and $k = 1$ (apart from the safe prime 5), and derive three further classical families from it: 2 is a primitive root modulo every prime $4\ell + 1$; 3 is a primitive root modulo every prime $16\ell + 1$; and 3 is a primitive root modulo every prime $8\ell + 1$ except 41. The development has seventeen main theorems in about 700 lines. We also describe a small checking script, stricter than a plain `lake build`, that rejects several ways of faking a successful build. We test it against deliberately tampered versions of the source. Version 2 adds two further checks. First, each of the seventeen theorems is restated with Mathlib-only types and proved, with an explicit non-degenerate witness. Second, an AI writer turned textbook statements of the results into Lean without seeing our code, and every one of its 26 statements was compared with the development. One of them, the general form of Wertheim's criterion as transcribed, is false as written; a corrected form is proved.

1 Introduction

An integer a is a *primitive root* modulo a prime p if its multiplicative order modulo p is exactly $p - 1$. Artin's conjecture on primitive roots (see Moree's survey [6]) predicts that every $a \neq -1$ that is not a perfect square is a primitive root for infinitely many primes. Remarkably, this is not known for any single a. Unconditional results come in two flavours: statements for sets of a (Heath-Brown [3]), and *explicit families*, where a special shape of $p - 1$ makes the multiplicative order easy to control.

This paper concerns two explicit families of the second kind:

1. **Chebyshev's criterion.** If p and $q = 4p + 1$ are both prime and $p \equiv 2 \pmod 5$, then 10 is a primitive root modulo q. The statement is quoted in [6, §1]. We formalise it together with its converse.

2. **Ramesh–Makeshwari.** If $p = 2q + 1$ is a *safe prime* (i.e. q is also prime), then the least positive primitive root modulo p is prime [7].

Neither result is deep. We formalise them because each depends on the same short piece of reasoning, and Mathlib now makes that reasoning inexpensive. There are three ingredients: the prime-divisor test for the order of a group element, Euler's criterion, and quadratic reciprocity. The formalisation also makes two small points explicit. One is the converse of Chebyshev's criterion. The other is that we package the standard existence theorem for primitive roots with the primality result, so the formal statement does not need a separate existence hypothesis.

Contributions.

- A Lean 4 formalisation of both results, with seventeen named theorems (Table 1), in the namespace `PrimitiveRootFamilies`.
- A general criterion for primitive roots modulo primes $2^k\ell + 1$ (Theorem 12), stated once for all $k \geq 1$, with the two families above as instances and three further families as corollaries (Section 5).
- A formal proof of the converse of Chebyshev's criterion (Theorem 3). It is elementary and very probably folklore. We do not claim it as new.
- A formal form of the Ramesh–Makeshwari theorem with no separate existence hypothesis (Theorem 10), with "least" expressed through Mathlib's `IsLeast`, so that the statement is about the least positive integer primitive root and not about a chosen representative.
- A proof-checking script ("gate") and a set of tampering tests, described in Section 7.
- (Version 2) A restatement of all seventeen theorems with Mathlib-only types, each with witnesses, and a comparison with independently written classical statements (Section 7.1).

Artifact. The development uses Lean 4 [5] and Mathlib [4]. The code is at `https://github.com/jpbald93/primitive-root-families`, commit `e30457a`. It uses Lean `v4.33.1` and Mathlib commit `0df444a3`, pinned in `lake-manifest.json`. Table 1 links each main declaration to that commit. The licence is Apache 2.0. Version 2 of this paper adds the files `PrimitiveRootFamilies/L14.lean` and `PrimitiveRootFamilies/L15Classical.lean` (Section 7.1) in a new commit, `b4ae30a94f5bc5bcba7e0143c239565af261038a`. The statements and contents of `Chebyshev.lean`, `General.lean` and `SafePrime.lean` are unchanged; version 2 updates the root import file, the gate's required-declaration list and the README. The links in Table 1 still point to commit `e30457a`. Supporting material (the Layer-14 and Layer-15 reports, the blind writer's statements and report, the transcript audit, and the reviews of this version with the fixes made in response) is in the folder `Primitive-root-families` of the repository `https://github.com/jpbald93/LEAN-verification`.

Lean name	Paper	File
`isPrimitiveRoot_iff_of_eq_four_mul_add_one`	Lemma 1	`Chebyshev.lean`
`legendreSym_ten_of_eq_four_mul_add_one`	Lemma 2	`Chebyshev.lean`
`isPrimitiveRoot_ten_iff_mod_five_eq_two`	Theorem 3	`Chebyshev.lean`
`isPrimitiveRoot_ten_of_mod_five_eq_two`	Corollary 4	`Chebyshev.lean`
`isPrimitiveRoot_iff_of_eq_two_mul_add_one`	Lemma 6	`SafePrime.lean`
`isPrimitiveRoot_of_legendreSym_eq_neg_one`	Lemma 7	`SafePrime.lean`
`prime_of_isLeast_isPrimitiveRoot`	Theorem 8	`SafePrime.lean`
`exists_isLeast_isPrimitiveRoot`	Lemma 9	`SafePrime.lean`
`exists_isLeast_isPrimitiveRoot_and_prime`	Theorem 10	`SafePrime.lean`
`isPrimitiveRoot_two_of_mod_four_eq_one`	Corollary 11	`SafePrime.lean`
`isPrimitiveRoot_iff_of_eq_two_pow_mul_add_one`	Theorem 12	`General.lean`
`isPrimitiveRoot_of_legendreSym_eq_neg_one_of_not_dvd`	Corollary 13	`General.lean`
`isPrimitiveRoot_two_of_eq_four_mul_add_one`	Corollary 14(a)	`General.lean`
`isPrimitiveRoot_three_of_eq_eight_mul_add_one`	Corollary 14(b)	`General.lean`
`isPrimitiveRoot_three_of_eq_sixteen_mul_add_one`	Corollary 14(c)	`General.lean`
`isPrimitiveRoot_iff_of_eq_two_mul_add_one'`	Theorem 12, $k = 1$	`General.lean`
`isPrimitiveRoot_iff_of_eq_four_mul_add_one'`	Theorem 12, $k = 2$	`General.lean`

Table 1: The seventeen main theorems and where they are stated. Each depends only on Lean's standard axioms `propext`, `Classical.choice` and `Quot.sound`.

2 Conventions and the Mathlib interface

Throughout, "a is a primitive root modulo n" is Mathlib's `IsPrimitiveRoot (a : ZMod n) (n - 1)`. By definition, $a^{n-1} = 1$, and $a^k = 1$ implies $(n-1) \mid k$. Equivalently (`IsPrimitiveRoot.iff_orderOf`), a has multiplicative order exactly $n-1$ in $\mathbb{Z}/n\mathbb{Z}$. For prime n this is the standard notion. Legendre symbols are Mathlib's `legendreSym p a`, for p prime and $a \in \mathbb{Z}$.

Everything rests on the following facts from Mathlib:

- `orderOf_eq_of_pow_and_pow_div_prime`:
  if $n > 0$, $x^n = 1$ and $x^{n/r} \neq 1$ for every prime $r \mid n$, then $\operatorname{ord}(x) = n$;
- `ZMod.pow_card_sub_one_eq_one` (Fermat's little theorem);
- `legendreSym.eq_pow` (Euler's criterion, for odd prime p): $\left(\frac{a}{p}\right) \equiv a^{(p-1)/2} \pmod{p}$;
- `legendreSym.at_two` (the second supplementary law);
- `legendreSym.quadratic_reciprocity_one_mod_four` (reciprocity when one prime is 1 mod 4);
- `HasEnoughRootsOfUnity.exists_primitiveRoot`, which provides a primitive $(p-1)$-th root of unity in $\mathbb{Z}/p\mathbb{Z}$, i.e. cyclicity of the unit group.

3 Chebyshev's base-10 family

Let p and $q = 4p + 1$ both be prime. Then $p \neq 2$, since $4 \cdot 2 + 1 = 9$ is composite. So p is odd and $q \equiv 5 \pmod{8}$.

Lemma 1 (Criterion for $q = 4p + 1$)**.** *Let p and $q = 4p + 1$ be prime and $a \not\equiv 0 \pmod q$. Then a is a primitive root modulo q if and only if $a^{2p} \not\equiv 1$ and $a^4 \not\equiv 1 \pmod q$.*

Proof. The order of a divides $q-1 = 4p$, whose prime divisors are 2 and p. Apply the prime-divisor test with $4p/2 = 2p$ and $4p/p = 4$. □

Lemma 2. *For primes p and $q = 4p + 1$, $\left(\frac{10}{q}\right) = -\left(\frac{q}{5}\right)$.*

Proof. Since $q \equiv 5 \pmod 8$, $\left(\frac{2}{q}\right) = -1$. Since $5 \equiv 1 \pmod 4$, reciprocity gives $\left(\frac{5}{q}\right) = \left(\frac{q}{5}\right)$. Multiply. □

Theorem 3 (Chebyshev's criterion and its converse)**.** *Let p and $q = 4p + 1$ be prime. Then 10 is a primitive root modulo q if and only if $p \equiv 2 \pmod 5$.*

Proof. Note first that $10 \not\equiv 0 \pmod q$, because $q \geq 13$. Also $10^4 \not\equiv 1 \pmod q$: otherwise $q \mid 9999 = 3^2 \cdot 11 \cdot 101$. The primes 3 and 11 are not of the form $4p+1$, and $101 = 4 \cdot 25 + 1$ with 25 composite. By Euler's criterion, $10^{2p} \equiv \left(\frac{10}{q}\right)$. Lemmas 1 and 2 then show that 10 is a primitive root if and only if $\left(\frac{q}{5}\right) = 1$, i.e. $q \equiv 1$ or $4 \pmod 5$. If $q \equiv 1 \pmod 5$ then $5 \mid p$, so $p = 5$ and $q = 21$, which is not prime. And $q = 4p + 1 \equiv 4 \pmod 5$ exactly when $p \equiv 2 \pmod 5$. □

Corollary 4 (Chebyshev)**.** *If p and $q = 4p+1$ are prime and $p \equiv 2 \pmod 5$, then 10 is a primitive root modulo q.*

The formal statement of the corollary is

```
theorem isPrimitiveRoot_ten_of_mod_five_eq_two {p q : ℕ} (hp : p.Prime) (hq : q.Prime)
    (hqp : q = 4 * p + 1) (hp5 : p % 5 = 2) : IsPrimitiveRoot (10 : ZMod q) (q - 1)
```

An earlier public version of the repository called this theorem `ten_isPrimitiveRoot`. That name remains available as a deprecated alias.

Remark 5 (On the converse). The "only if" direction of Theorem 3 uses nothing beyond the argument for the "if" direction. We searched for an explicit statement and found none. The underlying criterion is a standard textbook exercise: every quadratic nonresidue of a prime $4p+1$ is a primitive root or has order 4 (cf. Burton [1, §9.2, Problem 11(a)]). So we regard the converse as folklore and do not claim it as a new result.

Formalisation notes. The main source of friction is that Mathlib's reciprocity and supplementary laws are stated for natural-number primes cast to $\mathbb{Z}$. Rewriting the goal into the form `legendreSym q (5 : ℕ)` took more steps than the mathematics suggests. Mathlib deliberately provides no global instance `Fact (Nat.Prime 5)`, so we declare one locally. The step $q \nmid 9999$ is a four-way case split on the prime factors of 9999. Each case is closed by `omega`, except for 101, which needs the separate observation that 25 is not prime. The file ends with two `example`s: $p = 7$, $q = 29$, where 10 is a primitive root, and $p = 3$, $q = 13$, where it is not.

4 Safe primes

Now let $p = 2q + 1$ with p and q both prime. The argument has the same structure, with the exponent 4 replaced by 2.

Lemma 6 (Criterion for safe primes)**.** *If $a \not\equiv 0 \pmod p$, then a is a primitive root modulo p if and only if $a^q \not\equiv 1$ and $a^2 \not\equiv 1 \pmod p$.*

Lemma 7. *Every quadratic nonresidue x with $1 < x < p-1$ is a primitive root modulo p.*

Proof. By Euler's criterion $x^q \equiv \left(\frac{x}{p}\right) = -1 \not\equiv 1$. And $x^2 \equiv 1$ would force $x \equiv \pm 1 \pmod{p}$, which the range $1 < x < p-1$ excludes. □

Theorem 8 (Ramesh–Makeshwari [7])**.** *Let $p = 2q+1$ be a safe prime. If g is the least element of $\{a \in \mathbb{N} : a > 0,\ a \text{ is a primitive root modulo } p\}$, then g is prime.*

Proof. First, $g < p$. Otherwise $g \bmod p$ would be a smaller positive primitive root. It is positive because $p \nmid g$. By Lemma 6 and Euler's criterion, $\left(\frac{g}{p}\right) = -1$. Hence $g \neq 1$. Also $g \neq p-1$, since $(p-1)^2 \equiv 1$. Suppose $g = mn$ with $1 < m, n < g$. Then $\left(\frac{m}{p}\right)\left(\frac{n}{p}\right) = -1$, so one of m and n is a nonresidue in $(1, p-1)$. By Lemma 7, that factor is a primitive root smaller than g, a contradiction. □

Our proof is phrased with Legendre symbols. The published note [7] argues with multiplicative orders instead. The mathematical content is the same.

Existence. Theorem 8 is a statement about "the least g". It would be vacuously true if no positive primitive root existed. We therefore also prove existence.

Lemma 9. *For every prime p, the set of positive integers that are primitive roots modulo p has a least element.*

Proof. Mathlib provides a primitive $(p-1)$-th root of unity $\zeta \in \mathbb{Z}/p\mathbb{Z}$. Its canonical representative $\zeta.\mathrm{val} \in \{0, \ldots, p-1\}$ is nonzero and is a primitive root. The least element then exists by well-ordering (`Nat.find`). □

Theorem 10 (With existence)**.** *For every safe prime p, the least positive primitive root modulo p exists and is prime.*

```
theorem exists_isLeast_isPrimitiveRoot_and_prime {p q : ℕ} [Fact p.Prime] (hq : q.Prime)
    (hpq : p = 2 * q + 1) :
    ∃ g, IsLeast {a : ℕ | 0 < a ∧ IsPrimitiveRoot (a : ZMod p) (p - 1)} g ∧ g.Prime
```

Corollary 11. *If $p = 2q+1$ is a safe prime with $q \equiv 1 \pmod{4}$, then 2 is a primitive root modulo p.*

Proof. Here $q \geq 5$ and $p \equiv 3 \pmod{8}$, so $\left(\frac{2}{p}\right) = -1$. Apply Lemma 7. □

Formalisation notes. The only step that is not purely routine in the formal proof of Theorem 8 is the bound $g < p$. Informally, primitive roots are residues and nobody states it. In Lean the set is a set of natural numbers, and the bound must be derived from minimality. We use `Nat.exists_dvd_of_not_prime2` to obtain a nontrivial factorisation $g = mn$, and `legendreSym.mul` for multiplicativity. Existence (Lemma 9) takes ten lines, because Mathlib already has the cyclicity of $(\mathbb{Z}/p\mathbb{Z})^\times$ in the form of `HasEnoughRootsOfUnity`.

5 The general criterion for primes $2^k\ell + 1$

Lemmas 1 and 6 are, via Euler's criterion and apart from the safe prime $p = 5$, equivalent to the cases $k = 2$ and $k = 1$ of a single criterion. Throughout this section, ℓ is an odd prime, $k \geq 1$, and $p = 2^k\ell + 1$ is prime. Then $p - 1 = 2^k\ell$ has exactly the prime divisors 2 and ℓ.

Theorem 12. *Let $a \in \mathbb{Z}$ with $p \nmid a$. Then a is a primitive root modulo p if and only if $\left(\frac{a}{p}\right) = -1$ and $a^{2^k} \not\equiv 1 \pmod{p}$.*

Proof. By the prime-divisor test, a is a primitive root if and only if $a^{(p-1)/2} \not\equiv 1$ and $a^{(p-1)/\ell} = a^{2^k} \not\equiv 1$. By Euler's criterion, $a^{(p-1)/2} \equiv \left(\frac{a}{p}\right)$, and since $\left(\frac{a}{p}\right) = \pm 1$ with $p > 2$, the first condition is equivalent to $\left(\frac{a}{p}\right) = -1$. □

Corollary 13. *If $\left(\frac{a}{p}\right) = -1$ and $p \nmid a^{2^k} - 1$, then a is a primitive root modulo p.*

The formal version also assumes $a \not\equiv 0 \pmod{p}$ explicitly. This is redundant, since it follows from $\left(\frac{a}{p}\right) = -1$, but harmless.

The formal statement of Theorem 12 takes a in $\mathbb{Z}$ rather than in $\mathbb{Z}/p\mathbb{Z}$, so that the Legendre symbol can be stated directly:

```
theorem isPrimitiveRoot_iff_of_eq_two_pow_mul_add_one {p l k : ℕ} [Fact p.Prime] (hl : l.
    Prime)
    (hl2 : l ≠ 2) (hk : 1 ≤ k) (hpl : p = 2 ^ k * l + 1) {a : ℤ} (ha : (a : ZMod p) ≠
    0) :
    IsPrimitiveRoot (a : ZMod p) (p - 1) ↔
      legendreSym p a = -1 ∧ (a : ZMod p) ^ (2 ^ k) ≠ 1
```

Two short bridge theorems (Table 1) restate its cases $k = 1$ and $k = 2$ in this form. The case $k = 1$ needs the hypothesis $\ell \neq 2$, so it omits the safe prime $p = 5$; the case $k = 2$ does not, since $4 \cdot 2 + 1 = 9$ is composite. The Layer-15 file of version 2 covers $p = 5$ and the textbook form with $a \not\equiv -1$ in place of $a^2 \not\equiv 1$ (Section 7.1). Lemmas 1 and 6 were formalised first and are kept, with their original statements, for the proofs in the preceding sections.

Theorem 12 is the case $p-1 = 2^k\ell$ of a sufficient criterion that Dickson attributes to Wertheim [2, Vol. I, p. 196], together with its converse. As summarised there, the general form for $p - 1 = 2^k\ell^j$ is false when $j \geq 2$ (Section 7.1).

The criterion turns a primitive-root question into two checks: a Legendre symbol, handled by reciprocity, and finitely many exceptional primes, namely the prime divisors of $a^{2^k} - 1$.

Corollary 14. *Let ℓ be prime.*

(a) If $p = 4\ell + 1$ is prime, then 2 is a primitive root modulo p.

(b) If $p = 8\ell + 1$ is prime and $\ell \neq 5$, then 3 is a primitive root modulo p.

(c) If $p = 16\ell + 1$ is prime, then 3 is a primitive root modulo p.

Proof. (a) Here $\ell \neq 2$, since 9 is composite, so $p \equiv 5 \pmod{8}$ and $\left(\frac{2}{p}\right) = -1$. If $2^4 \equiv 1$ then $p \mid 15$, which is impossible since $p \geq 13$.

(b) The case $\ell = 2$, $p = 17$, is checked directly. Otherwise ℓ is odd and $p \equiv 1 \pmod{4}$, so $\left(\frac{3}{p}\right) = \left(\frac{p}{3}\right)$. Now $p \equiv 2\ell + 1 \pmod{3}$: $\ell \equiv 1$ gives $3 \mid p$, and $\ell = 3$ gives $p = 25$. Hence $p \equiv 2 \pmod{3}$ and $\left(\frac{3}{p}\right) = -1$. If $3^8 \equiv 1$, then $p \mid 6560 = 2^5 \cdot 5 \cdot 41$, so $p = 41$ and $\ell = 5$, which is excluded.

(c) Here $\ell \neq 2$, since 33 is composite, and $p \equiv \ell + 1 \pmod 3$: $\ell \equiv 2$ gives $3 \mid p$, and $\ell = 3$ gives $p = 49$. So $\left(\frac{3}{p}\right) = -1$ as in (b). If $3^{16} \equiv 1$ then $p \mid 3^{16} - 1 = 2^6 \cdot 5 \cdot 17 \cdot 41 \cdot 193$, and none of $5, 17, 41, 193$ is of the form $16\ell + 1$ with ℓ prime. □

The exception in (b) is genuine: $3^8 \equiv 1 \pmod{41}$, so 3 is not a primitive root modulo $41 = 8 \cdot 5 + 1$. The formal file proves this as an `example`. In fact its order is 8, since $3^4 \equiv -1 \pmod{41}$. It illustrates the second of the two checks: a nonresidue can fail to be a primitive root only at a prime divisor of $a^{2^k} - 1$. Corollary 11 is a relative of part (a): it applies the general criterion with $a = 2$, $k = 1$ and $\ell \equiv 1 \pmod 4$. Part (a) is Burton's §9.2, Problem 11(b) [1]. Parts (b) and (c) are elementary applications of the same method, and we make no novelty claim for any of the three.

6 Numerical cross-checks

The repository contains three short Python scripts. They are independent of the Lean development and serve as a sanity check that the formal statements say what we intend.

- `code/check_family.py 200000`: there are 1916 prime pairs $(p, 4p + 1)$ with $p \leq 2 \cdot 10^5$. In every case, 10 is a primitive root modulo $4p + 1$ exactly when $p \equiv 2 \pmod 5$. That happens for 628 of the pairs. As a control, for $p = 22$ (not prime) and $q = 89$, the order of 10 is 44.
- `code/check_safe_primes.py 1000000`: there are 4324 safe primes $p \leq 10^6$. For every one, the least primitive root is prime. For the 2133 with $q \equiv 1 \pmod 4$, 2 is a primitive root. For the 72 safe primes below 5000, a brute-force computation of orders confirms that all 73514 nonresidues in $(1, p-1)$ are primitive roots. As a control, the least primitive root of the prime 41, which is not a safe prime, is the composite number 6.
- `code/check_general.py`: for the 16563 primes $p = 2^k\ell + 1 \leq 2 \cdot 10^6$ with ℓ an odd prime, Theorem 12 agrees with SymPy's primitive-root test in all 182191 cases (p, a) tested, for the bases $a \in \{2, 3, 5, 6, 7, 10, 11, 12, -1, -2, -3\}$. SymPy's test uses the same prime-divisor method, so as an independent control, orders are also computed by brute force in the 1219 cases with $p < 3000$. In the same range, 2 is a primitive root modulo all 4109 primes $4\ell + 1$, and 3 is a primitive root modulo all 1181 primes $16\ell + 1$. Among the 2158 primes $8\ell + 1$, the only failure for 3 is $p = 41$.

For interest, Table 2 gives the distribution of the least primitive root over the same 4324 safe primes. The least primitive root is 2 for the 2133 primes with $p \equiv 3 \pmod 8$ and also for $p = 5$. The value 3 occurs only for $p = 7$. This agrees with a one-line computation, not formalised here. For a safe prime $p > 7$, we have $p \equiv 3 \pmod 4$ and $p \equiv 2 \pmod 3$, so reciprocity gives $\left(\frac{3}{p}\right) = -\left(\frac{p}{3}\right) = +1$. Hence 3 is never a primitive root modulo such a p.

g	2	3	5	7	11	13	17	19	23	29	31	37	41
count	2134	1	1461	290	194	141	54	24	15	3	3	3	1

Table 2: Least primitive root g of the 4324 safe primes $p \leq 10^6$.

7 A tamper-resistant check

A `lake build` that succeeds does not by itself show that a statement has been proved. The source may contain `sorry`, a user-declared `axiom`, or `native_decide`. It may also contain code that runs

at elaboration time and prints misleading output. For example, `#eval` can print text that looks like a clean `#print axioms` report, and `macro_rules` can redefine what `#print axioms` does. The development was prepared with AI assistance (see Acknowledgements), so we treated these as real risks. The repository's `gate.sh` therefore does the following:

1. It rejects any source file that contains `sorry`, `admit`, `native_decide`, the `axiom` keyword, `#eval`, `run_cmd`, `initialize`, `IO`, `set_option`, or syntax-extension commands (`macro`, `elab`, `syntax`, `notation`, `import Lean`, ...).
2. It builds the library.
3. It writes its own temporary file of `#print axioms` queries, one for each theorem in Table 1, instead of trusting any report produced inside the repository.
4. It flattens Lean's line-wrapped output, requires exactly one report per theorem and no others, and accepts only the axioms `propext`, `Classical.choice` and `Quot.sound`.

We tested the gate by planting each of the following in a copy of the repository and confirming that the gate fails: a `sorry`; an `axiom` whose keyword sits on a line of its own; a fake clean report printed with `#eval`; a `macro_rules` redefinition of `#print axioms`; an extra axiom hidden on a wrapped output line (with the source filter's `axiom` check switched off, so that the output parser is tested alone); and a false hypothesis ($p \bmod 5 = 3$ in place of 2), which breaks the proof. The unmodified repository passes with `PASS (17 theorems, standard axioms only)`. The script `tests/tamper.sh` in the artifact reproduces all six tests: each one plants its defect in a scratch copy of the repository and checks that the gate fails at the expected stage.

7.1 Layers 14 and 15

Version 2 of the artifact adds two more checks, called Layers 14 and 15. Both were carried out by AI agents. The library files `Chebyshev.lean`, `General.lean` and `SafePrime.lean` are unchanged: their hashes were compared before and after. Version 2 updates the root import file and the gate's required-declaration list.

Layer 14: Mathlib-only statements and witnesses. The file `PrimitiveRootFamilies/L14.lean` (376 lines) restates all seventeen theorems of Table 1. No theorem is omitted. The published statements already mention only Mathlib constants, so each restatement goes further and uses textbook phrasings. Primality is an explicit hypothesis `Nat.Prime p`, not an instance. "Primitive root" is `orderOf a = p - 1`, via `IsPrimitiveRoot.iff_orderOf`. The Legendre symbol is `jacobiSym a p`, via `jacobiSym.legendreSym.to_jacobiSym`. Each restatement `l14_*` is proved from the library theorem and these bridges, with the same hypotheses. A mechanical check of the elaborated types found no `PrimitiveRootFamilies` constant in any of the 36 new declarations.

Each restatement also has a witness theorem `l14w_*`, which proves every hypothesis on an explicit instance together with a non-degenerate conclusion. For each equivalence, the witness exhibits both truth values. Two of them show that the condition $a^{2^k} \not\equiv 1$ in Theorem 12 is needed: $a = 3$ for $p = 41$, and $a = 5$ for $p = 13$. In each case a is a nonresidue but not a primitive root. For Theorem 8 the hypothesis `IsLeast` is realised explicitly: the least primitive root of 23 is 5. For Corollary 14(b) the witness uses the generic case $p = 89$, not just $p = 17$, and also proves that 3 is not a primitive root modulo 41. No Mathlib-only statement turned out to be false or to need an extra hypothesis. With Layer 14 the gate passes with `PASS (53 theorems, standard axioms`

`only)`, and the tampering tests still reject every planted defect. One test (`wrap`) replaces a library theorem by `True`, so a downstream file that used that theorem stopped building and the test failed at the wrong stage. The restatement of Corollary 4 is therefore derived from Theorem 3.

Layer 15: comparison with classical statements. A separate agent collected the classical statements from the literature, with citations: Burton, Hardy–Wright, Dickson's *History* [2], the Ramesh–Makeshwari note [7], and encyclopaedic sources. They fall into six groups: Euler's criterion; the primitive-root test; the Fermat-prime, safe-prime and Wertheim characterisations; Chebyshev's theorems on 2 and on 10; and Ramesh–Makeshwari, with the lemma that the least nonresidue is prime. An AI writer received only these statements and saw no code and no paper. Its transcript was audited, and its report records that it opened no other file. It wrote 26 Lean statements with Mathlib notions, including two existence obligations, all with `sorry`. A comparer preserved their hypotheses and conclusions, repackaging the theorem declarations as universally quantified proposition definitions, and proved each one from the development and Mathlib, or refuted it. No hypothesis was added and the library was not changed. All 28 resulting theorems depend only on the three standard axioms. The comparison was first checked as a scratch file; it was then integrated as `PrimitiveRootFamilies/L15Classical.lean` (704 lines), in the namespace `PrimitiveRootFamilies.Classical`, with the definitions and proposition statements byte-identical to the checked comparison file and every proof kept. With both layers the gate passes with `PASS (83 theorems, standard axioms only)`, and all six tampering tests are still rejected. The outcome for the 26 statements:

- *Covered* (8): the statements on 2 modulo $4q+1$ (the library version does not need $q \neq 2$), the three forms of Chebyshev's base-10 criterion, its examples $29, 149, 269$, the note that p is odd and $q \equiv 5 \pmod 8$, and Ramesh–Makeshwari with its existence obligation.

- *Narrower* (1): the safe-prime criterion "primitive root $\iff \left(\frac{a}{p}\right) = -1$ and $a \not\equiv -1$". The library has it only for $\ell \neq 2$ and with $a^2 \not\equiv 1$, so the case $p = 5$ and the translation to $a \not\equiv -1$ were missing. Both are supplied by a new proof, so the statement is now proved in full.

- *Out of scope* (16): background results that the paper uses from Mathlib or does not claim. These are Euler's criterion (3 statements), the primitive-root test and its corollaries (4), Fermat primes, two remarks on -1 for safe primes, the decimal-period form of Chebyshev's criterion, the examples of least primitive roots, the lemma that the least nonresidue is prime and exists (2), and "least primitive root = least nonresidue" for safe primes with its bound (2). All 16 were proved.

- *Writer statement false as written* (1): Wertheim's criterion for $p = 2^k q^l + 1$ with $k, l \geq 1$. It is false for $l \geq 2$: for $p = 19 = 2 \cdot 3^2 + 1$ and $a = 8$ we have $\left(\frac{8}{19}\right) = -1$ and $8^2 \not\equiv 1$, but 8 has order 6. The writer transcribed the source statement faithfully, so the error lies in the source. Theorem 12 is its case $l = 1$, which is correct and is derived from the library. A corrected criterion, with $a^{2^k q^{l-1}} \not\equiv 1$, is proved.

These proofs were absent from the version-1 library and are now included in the Layer-15 file:

- `cov_Prime3_b`: the safe-prime criterion including $p = 5$, with the condition $a \not\equiv -1$.

- `not_classical_Prime3_c`: the counterexample $p = 19$, $a = 8$.

- `new_Prime3_c_corrected`: Wertheim's criterion for general l, corrected.

- `cov_Prime3_a`: for a Fermat prime, the primitive roots are exactly the nonresidues.
- `cov_Prime5_reptend`: for primes p and $q = 4p + 1$, 10 is a primitive root modulo q iff the decimal period of $1/q$ is $q - 1$, using `decimalPeriod_eq_orderOf`, which proves the period/order equality for prime q with $q \nmid 10$.
- `cov_Prime6_a`, `cov_Prime6_a_obligation`: the least positive nonresidue exists and is prime.
- `cov_Prime6_b_aux`, `cov_Prime6_b`: for a safe prime, the least nonresidue is below $p - 1$ and equals the least primitive root.
- thin wrappers around Mathlib for Euler's criterion and the primitive-root test.

Limitations. The token filter is a heuristic over the project's own sources, and it is deliberately over-strict. The gate trusts the pinned toolchain, Mathlib and lake configuration. At commit `e30457a` it audits only the seventeen named theorems; with Layer 14 it audits 53 declarations, and with Layers 14 and 15 together 83. It is not a sandbox. It also does not compare theorem statements against a trusted specification. Layer 15 compares them with independently written classical statements, but that comparison was done by AI agents. That the formal statements mean what the paper says still requires a human reading, which Sections 3–5 are intended to support. Environment replay with `leanchecker` (formerly `lean4checker`) rechecks compiled declarations using Lean's own kernel, and independently implemented external checkers provide a different trust boundary. Our gate is a lightweight complement to these tools, not a substitute.

8 Related work

Mathlib contains the general theory used here: primitive roots, the cyclicity of $(\mathbb{Z}/p\mathbb{Z})^\times$, Euler's criterion and quadratic reciprocity. We found neither family in the pinned Mathlib version. We found no prior formalisation of either result in Lean, Isabelle, Coq/Rocq or Mizar, but our search was not exhaustive. The broader context, Artin's conjecture and its partial results, is surveyed by Moree [6]. Other formalisations of elementary number theory include the Isabelle AFP and Mathlib's `NumberTheory` library.

9 Conclusion

Chebyshev's criterion, with its converse, and the Ramesh–Makeshwari theorem both follow from one prime-divisor criterion together with Euler's criterion. With current Mathlib, each can be formalised in about 220 lines, and the criterion itself holds uniformly for primes $2^k\ell+1$ (Section 5). Formalising them made two points explicit: the converse of Chebyshev's criterion, and the existence of the least root. Natural extensions are primes p with $p-1 = 2^k\ell^j$ or with more prime factors, where the exponents change and each additional distinct prime factor adds a condition. The sufficient direction for $2^k\ell^j$, with the condition $a^{2^k\ell^{j-1}} \not\equiv 1$, is proved in the Layer-15 file; the converse and further prime factors remain. Other extensions are criteria that also use cubic reciprocity (Fueter's results, surveyed in [6, §9.7, item 33]), and a formal proof of the observation in Table 2 about the least primitive root.

Acknowledgements and use of AI

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